

Big Data Modeling & Analytics

10 Weeks Course Outline

Mahmoud Parsian

Ph.D. in Computer Science



Big Data Modeling & Analytics:

1. What is Big Data?
2. What is Big data Modeling?
3. What is Big Data Modeling & Analytics?
4. **Solutions** to Big Data Modeling & Analytics
 - **1. MapReduce paradigm**
 - **2. Spark/PySpark**
 - **3. Amazon Athena & Google's BigQuery**



1. What is Big Data?

Big Data refers to datasets that are so massive, fast-moving, and complex that traditional data-processing software **cannot** capture, store, or analyze them effectively.

Instead of relying on a single computer, big data requires cluster computing frameworks like

- Apache Spark
- Snowflake
- Amazon Athena
- Google's BigQuery



What is a Big Data Modeling?

Big data modeling is the process of **organizing large**, **fast-moving**, and **diverse datasets** so they can be **stored**, **processed**, **analyzed**, and **understood** efficiently.

Key Characteristics of Big Data:

1. **Volume**: Handles massive amounts of data
2. **Velocity**: Supports rapidly generated and processed data
3. **Variety**: Combines structured, semi-structured, and unstructured data
4. **Veracity**: The quality, reliability, and trust of the data
5. **Scalability**: Adapts as data and user demand grow



What is a Big Data Modeling?

Common Models

- Data lakes and Lake-Houses
- NoSQL models: document, key-value, column, and graph
- Distributed data warehouses (Snowflake)
- Batch and streaming architectures (Spark)

> **Goal:** Turn complex raw data into a reliable structure for analysis and decision-making.



What is a Big Data Modeling & Analytics?

Big data analytics uses modeled data to discover patterns, explain outcomes, predict future events, and recommend actions.

Analytics Types

- **Descriptive**: What happened?
- **Diagnostic**: Why did it happen?
- **Predictive**: What is likely to happen?
- **Prescriptive**: What action should be taken?

Typical Workflow

Collect → Clean → Model → Process → Analyze → Visualize → Act



Business Value of Big Data Modeling & Analytics?

1. Faster, evidence-based decisions (like DNA analysis)
 2. Improved customer experiences (like Amazon)
 3. Fraud and risk detection (Credit-Cards like VISA)
 4. Operational efficiency (customer services)
 5. Forecasting and innovation (Weather data)
 6. Building AI-LLM models (like Claude, OpenAI)
 7. Building Search Engines (like Google)
-
- Strong analytics depends on a scalable, accurate,
 - and well-governed data model.





Query-Driven Schema Design for Big Data

Unlike traditional relational modeling — where the schema is normalized first and queries are written against it afterward — big data modeling typically works in the **opposite order**:

1. Study the Requirements. Identify who needs the data and why: which dashboards, reports, ML features, or applications will consume it.

2. Enumerate the Access Patterns. Determine exactly how the data will be queried — which fields are filtered on, which are aggregated, how the data is expected to grow, and how fresh it needs to be.

3. Design the Schema to Fit those Queries. Choose partition keys, denormalization strategy, and physical layout so the known queries run fast — even if that means duplicating data or shaping the same entity differently across multiple tables.



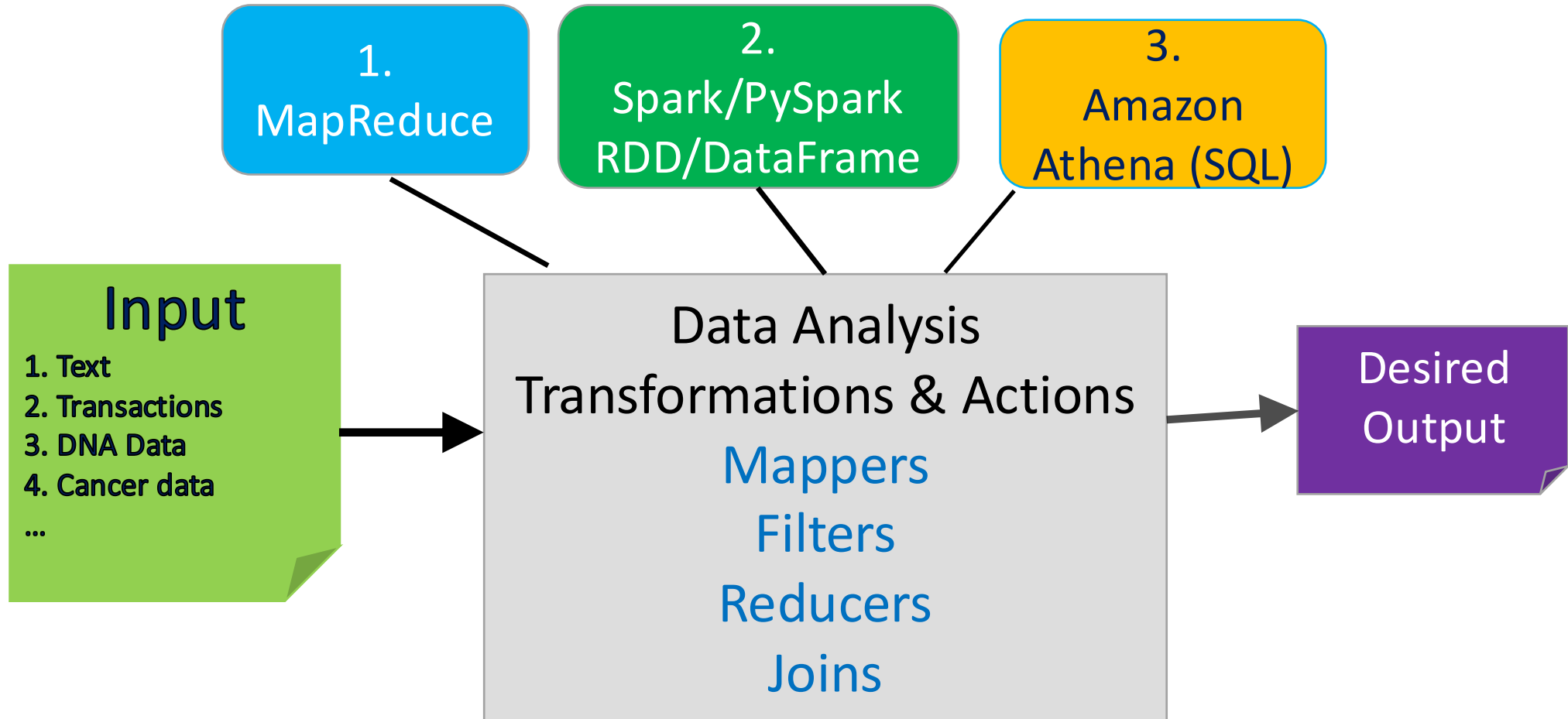
Big Data Modeling

Course Components:

1. Introduction to Big Data (5%)
2. Introduction to MapReduce Paradigm (20%)
3. Spark and PySpark (60%)
4. Serverless SQL Access to Big Data (15%)



Big Data Analysis



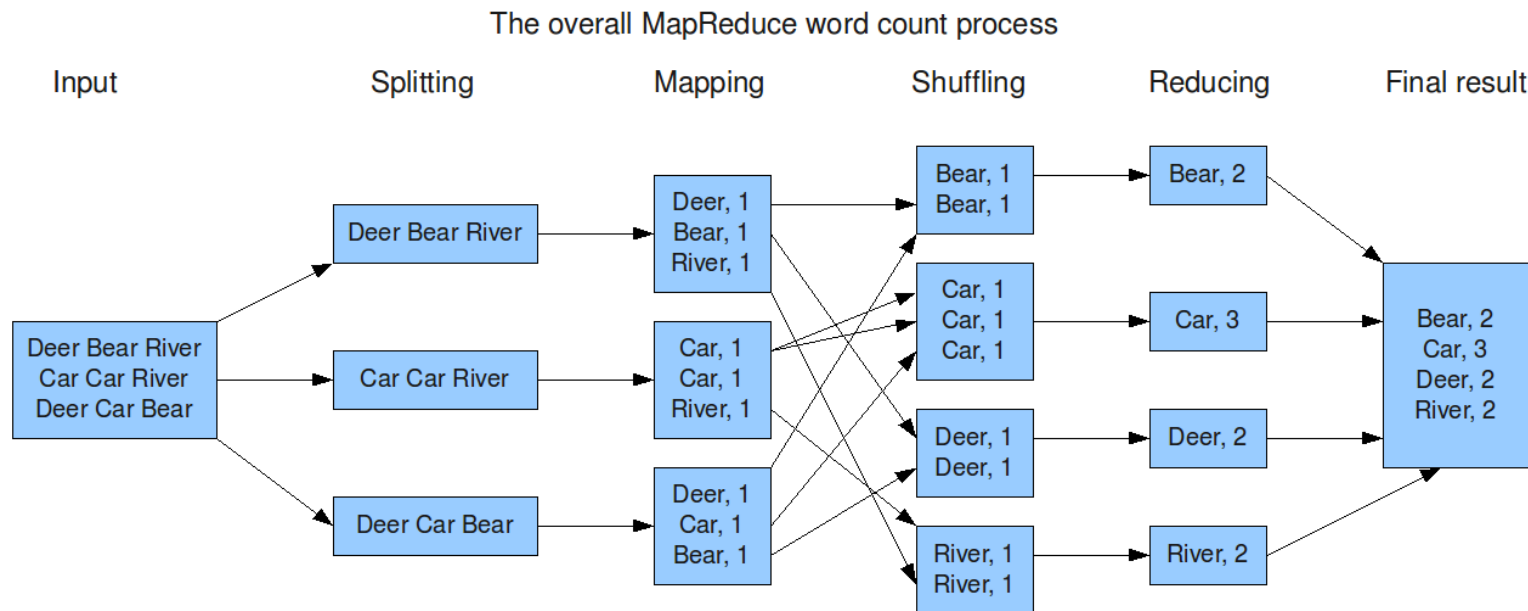
1. MapReduce Paradigm

- **MapReduce** is a programming **paradigm/model** that enables **massive scalability** across hundreds or thousands of servers in a cluster.
- **MapReduce has 3 functions:**
 - `map()`
 - `reduce()`
 - `combine()` [optional function]



1. MapReduce Paradigm in Action:

split data into chunks: then parallelize



1. MapReduce Programs

- **For MapReduce, we will use:**
 - Pseudo code (not executable)
 - PySpark code (runs on Spark)
- **For MapReduce, we will focus on learning MapReduce paradigm/model that enables massive scalability across hundreds or thousands of servers in a cluster.**
- **For MapReduce, pseudo-code**
 - examples are given in Lin, J., & Dyer, C. (2010). *Data-intensive text processing with MapReduce*



1. MapReduce Components

MapReduce:

- split data into chunks and then
- use 3 functions and execute tasks in parallel:

map()

- Mapper function

reduce()

- Reducer function

combine()

- [Optional combiner function]

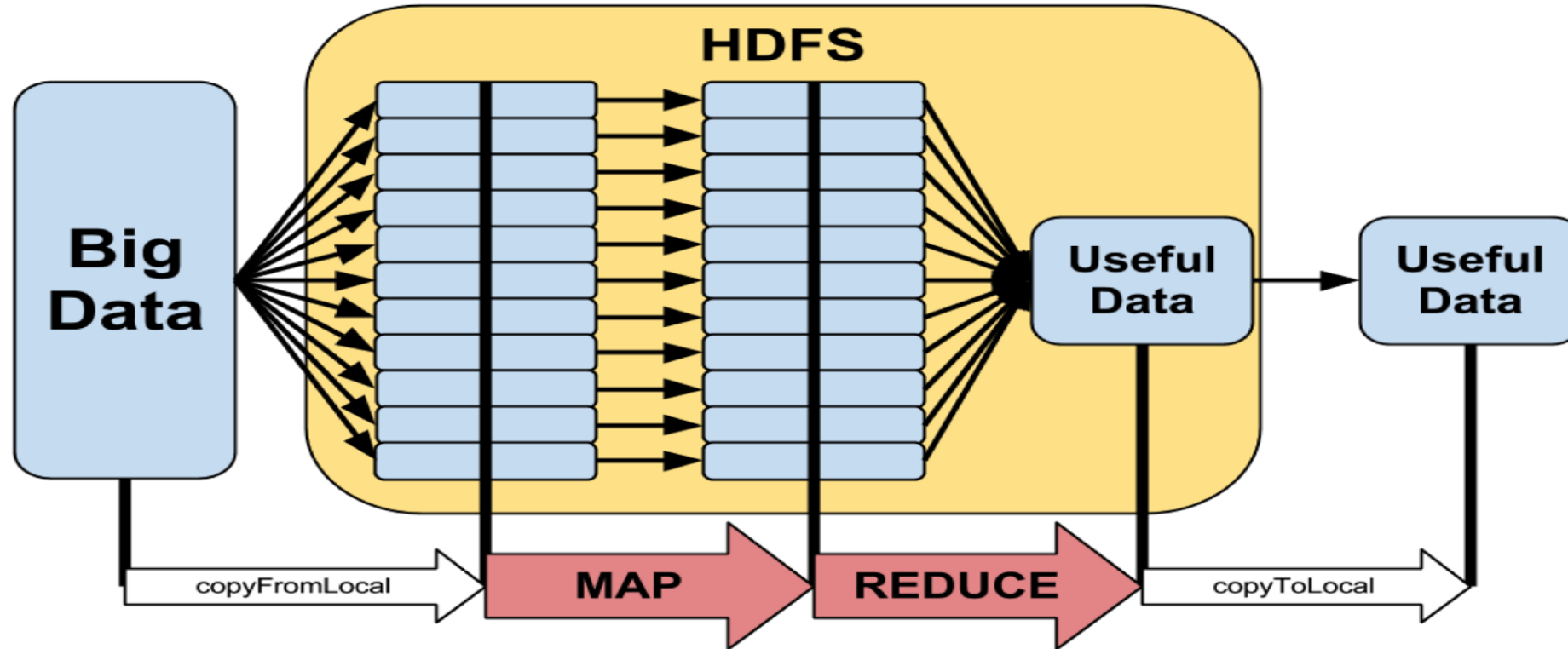


MapReduce Paradigm

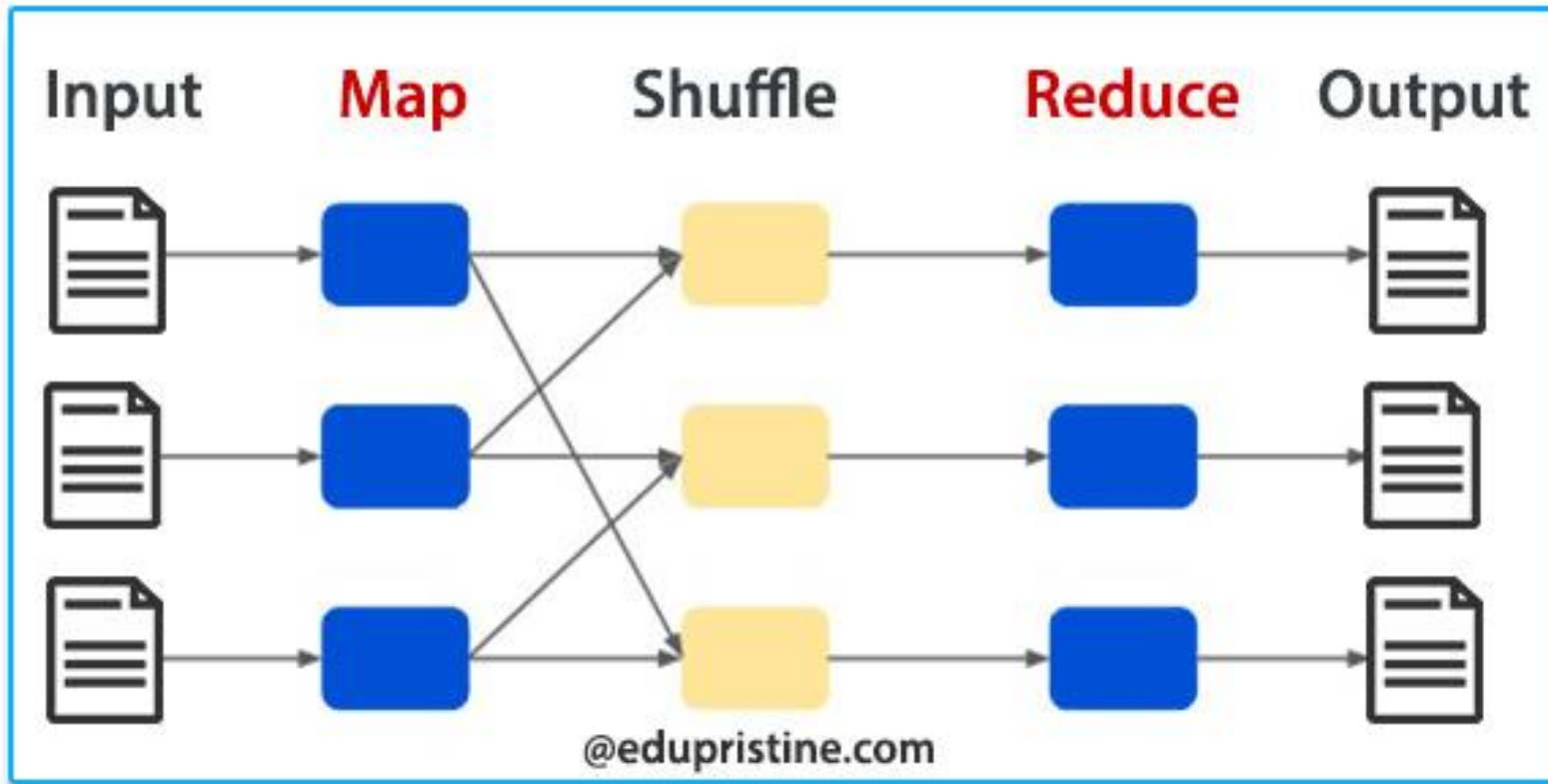
- MapReduce is a foundation model/paradigm for distributed computing using clusters
- MapReduce Implementations:
 - Apache Hadoop implements MapReduce
 - Apache Spark implements superset of MapReduce
 - Apache Tez implements MapReduce
 - Amazon Athena: Serverless SQL service
 - Google BigQuery: distributed execution graph and shuffle architecture



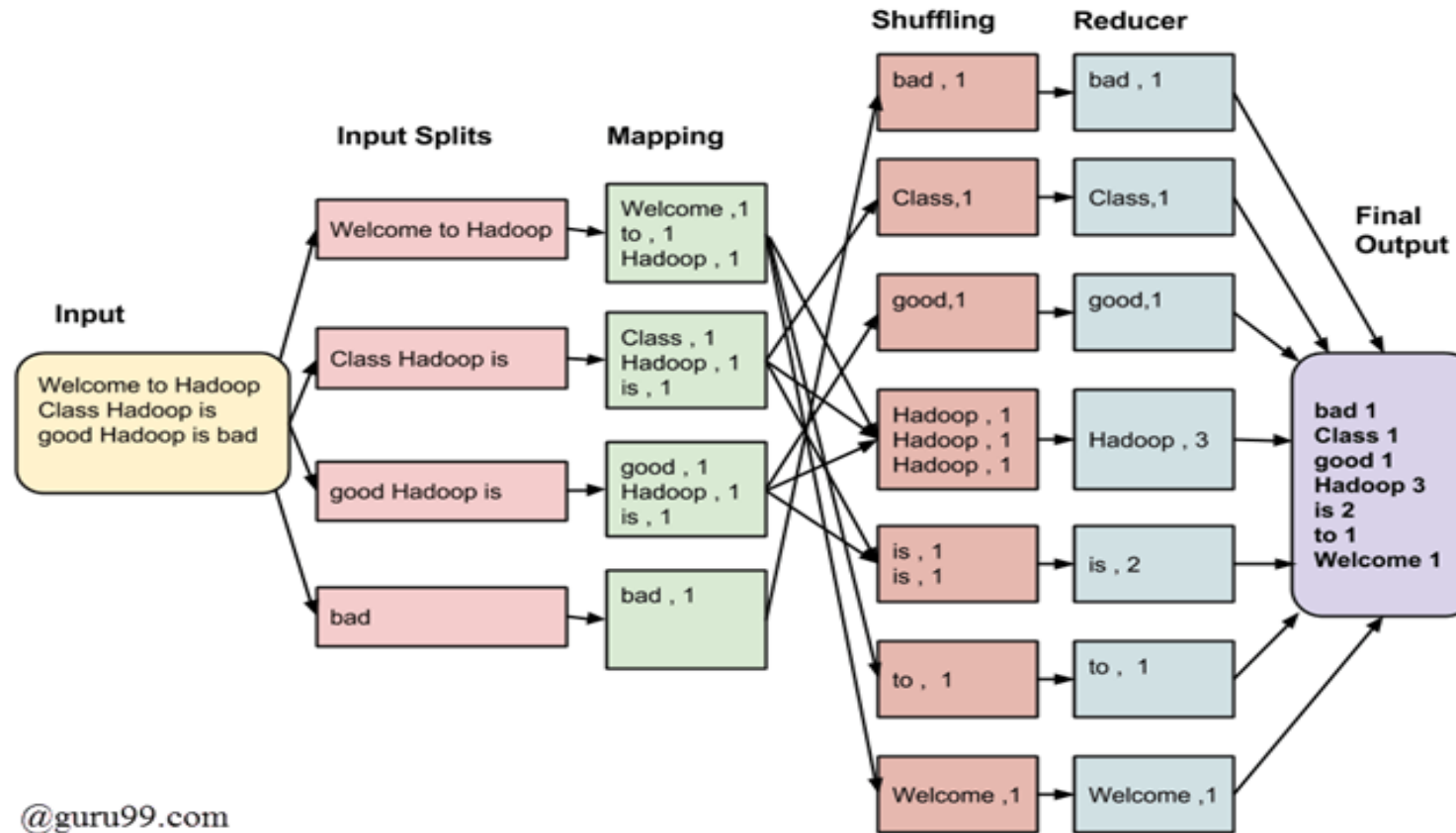
MapReduce Paradigm: High-Level



MapReduce Paradigm



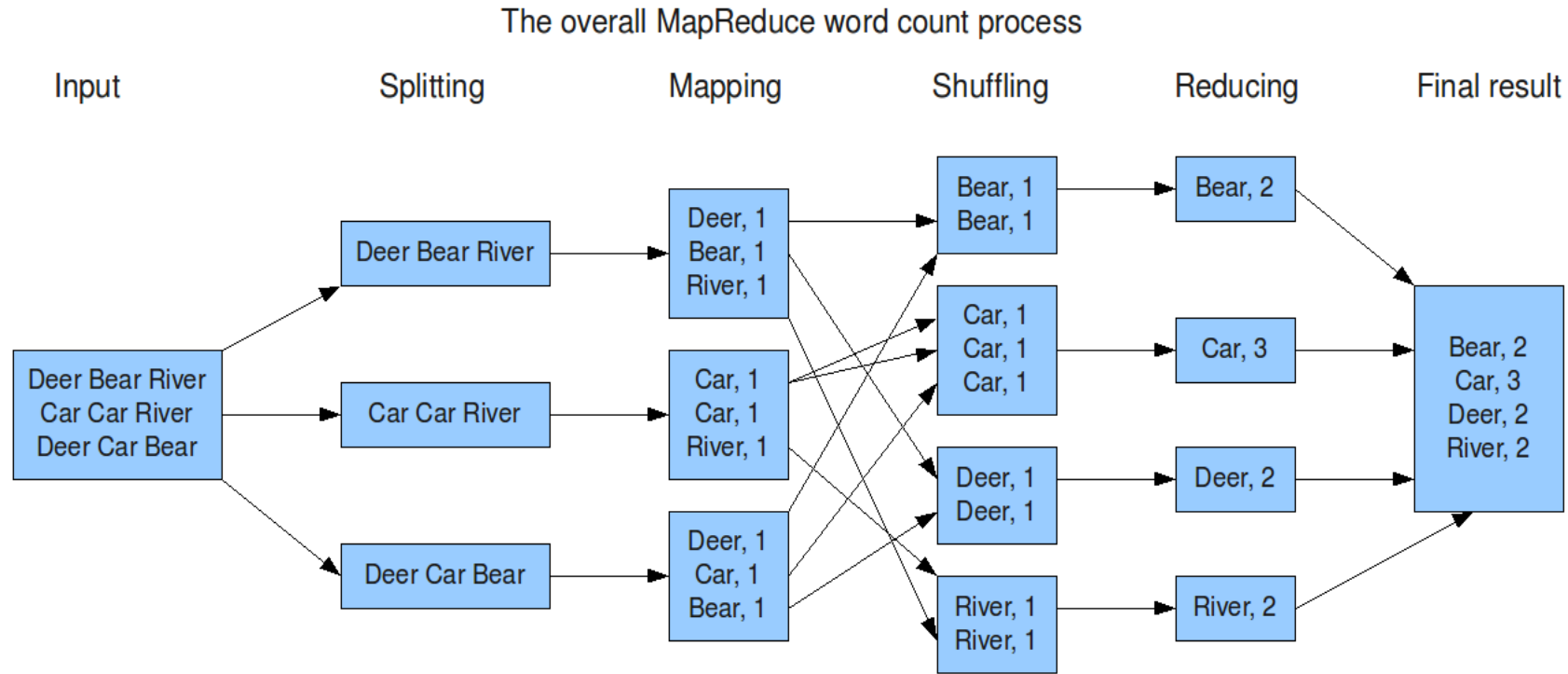
MapReduce Paradigm Example-1



@guru99.com



MapReduce Paradigm Example-2



Solving a Big Data Problem with MapReduce (1)

- Input data is partitioned by a partitioner into chunks and sent to Mappers
- Assume your input is comprised of 80,000,000,000 records, which might be partitioned into 40,000 chunks (partitions)
 - Number of partitions: 40,000
 - Each partition will have about 2,000,000 records
 - Maximum mapper parallelism can be 40,000 mappers
 - $80,000,000,000 = 40,000 \times 2,000,000$
- A mapper function, $\text{map}(K, V)$ is executed on all partitions in parallel (as much as possible – depends on resources available).
- The $\text{map}(K, V)$ generates any number of (key, value) pairs such as:
- $\{ (K1, V1), (K2, V2), \dots \}$. For example, K can be a record number and V can be the actual input record



Solving a Big Data Problem with MapReduce (2)

- All mappers output go to Sort & Shuffle phase, which groups values by unique keys (similar to GROUP BY in SQL).
- **Sort & Shuffle** creates (key, value) pairs as:

(**key_1**, [v1, v2, ...])

(**key_2**, [u1, u2, ...])

...

(**key_n**, [t1, t2, ...])

Where {**key_1**, **key_2**, ... **key_n** }

are the unique keys created by all mappers



Solving a Big Data Problem with MapReduce (3)

- Output of Sort & Shuffle goes to reducers
- A reducer operates on (key, value) pairs where key is in

`{ key_1, key_2, ..., key_n }`

and value is an associated value for the key.

Therefore, the following reducers can be executed in parallel:

`reduce(key_1, [v1, v2, ...])`

`reduce(key_2, [u1, u2, ...])`

`...`

`reduce(key_n, [t1, t2, ...])`

- Each reducer can create any number of new (key, value) pairs.



2. **Apache Spark:** engine for large-scale data analytics

- **Spark** is a multi-language (Java, Python, Scala) engine for executing data engineering, data science, and machine learning on single-node machines or clusters.
- **PySpark** is a Python API for Spark
- **Spark** is a superset of MapReduce
- **Spark** web site: <https://spark.apache.org>



2. Apache Spark Programs

For Apache **Spark** , we will use PySpark and write **actual executable programs**:

```
>>> # spark is an instance of SparkSession object
```

```
>>> rdd = spark.sparkContext
```

```
                .parallelize(range(0, 100))
```

```
>>> rdd.count()
```

```
100
```

```
>>> sum_of_values = rdd.reduce(lambda x, y: x+y)
```

```
>>> sum_of_values
```

```
4950
```



Apache Spark: Components

Streaming

MLlib

For Machine Learning

GraphX

For Graph Computing

**Spark SQL &
DataFrames**

Spark Core API

R

Python

Scala

SQL

Java



Spark Run Modes

Local Mode Single machine:

- Runs entirely on a single computer or laptop.
- Threads for execution: Uses local threads to simulate worker nodes.
- Testing and development: Best used for debugging and running quick tests.

Cluster Modes:

In cluster modes, the main distinction is where the Spark Driver (the program that coordinates the work) runs.

- **Client Mode:** External driver: The driver runs on the machine where the job was submitted. Interactive use: Best for interactive shells, notebooks, and local debugging. Network dependent: If your local machine disconnects, the job fails.
- **Cluster Mode:** Internal driver: The driver runs inside a worker node within the cluster. Production standard: Best for scheduled production jobs. Independent: You can close your laptop and the job will keep running.



Spark Run Modes: Local Single Machine or Cluster

The code that runs on a single machine is the same code which runs it in a cluster of 100's of nodes: In Spark, this concept is expressed as:

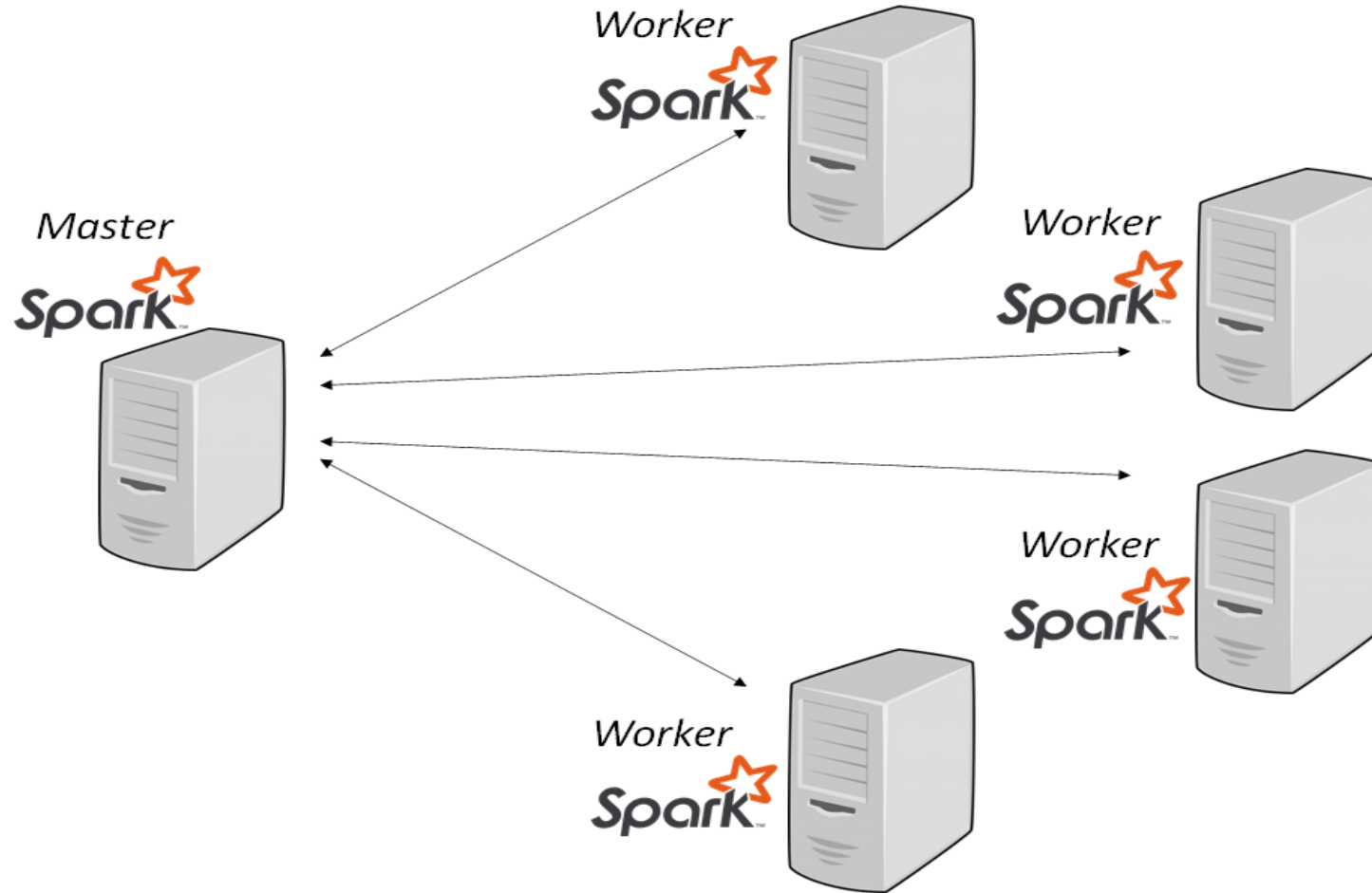
"The exact same PySpark code runs seamlessly on a single machine or scales out to a cluster of 100s of nodes without changing a single line of logic."

Write Once. Run at Any Scale.

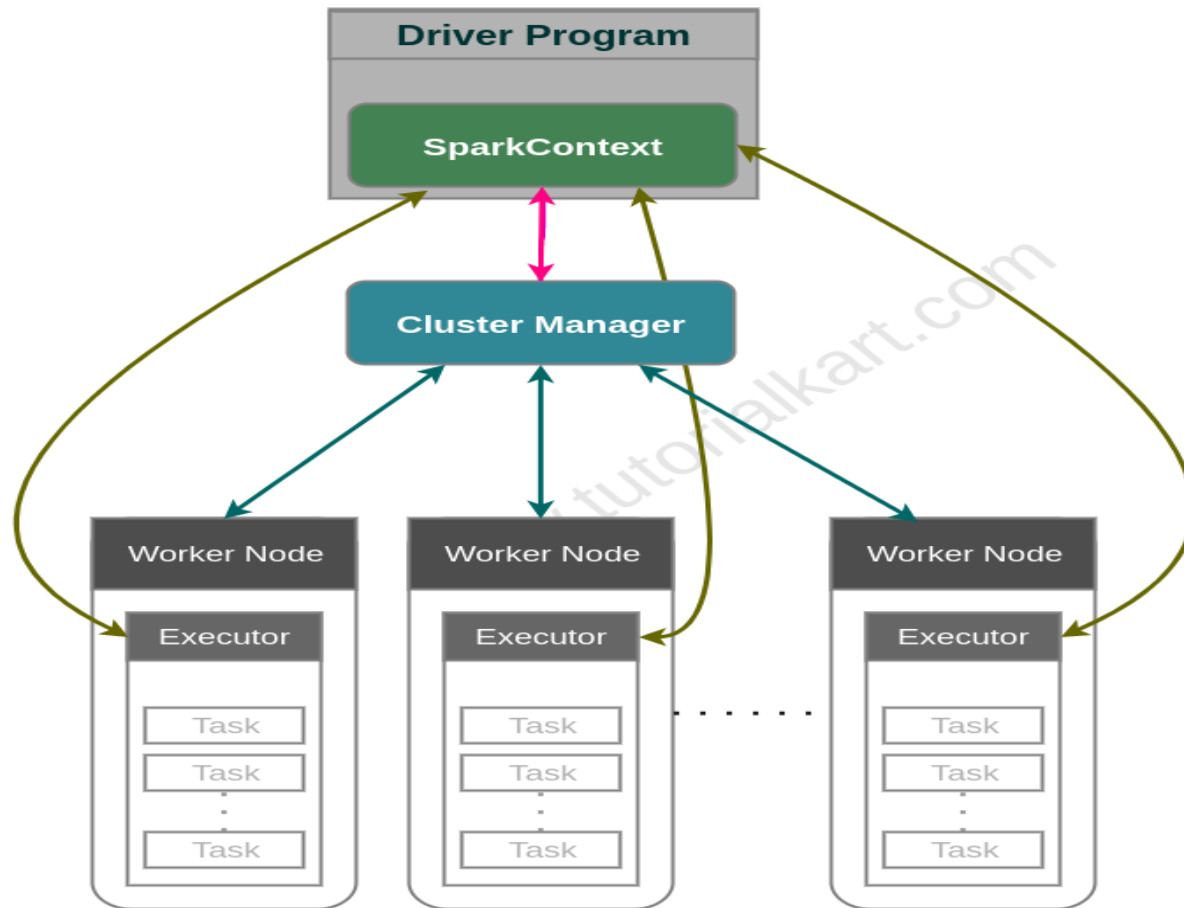
1. The **same code** runs on a single machine or a cluster of 100s of nodes.
2. Develop locally: Build and test on one machine.
3. Scale through configuration: Choose the cluster and resources.
4. Let the engine distribute the work: It coordinates parallel execution across nodes.
5. With a distributed processing engine, the execution environment changes—the application code can stay the same.



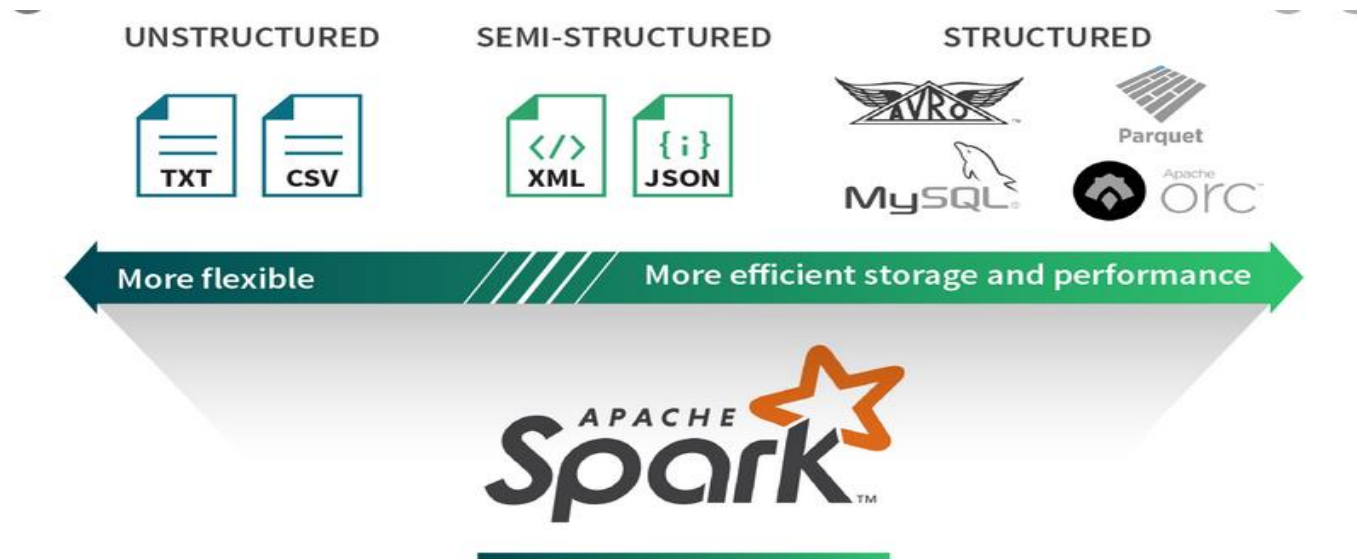
Apache Spark: runs in a cluster



Apache Spark: Cluster Manager



Spark Data Abstractions: Read/Write from/to Many Data Sources



Spark Data Abstractions

Your data can be represented as an **RDD** and **DataFrame**

1. Resilient Distributed Datasets (RDD)

- Data is represented as a data type T (**semi-structured data**)
(integer, string, tuples, arrays, dictionary, ...)
- Billions of data points (elements/records)

2. DataFrame

- Data is represented as a **table of rows and named columns** (**structured data**)
- Billions of rows of data



Spark Data Abstractions: RDDs

Billions of data elements

Each RDD element: `(String, (Float, Float))`

Element-1:

`(gene-1, (3.0, 4.5))`

Element-2:

`(gene-2, (1.0, 1.5))`

Element-3:

`(gene-1, (2.1, 1.6))`

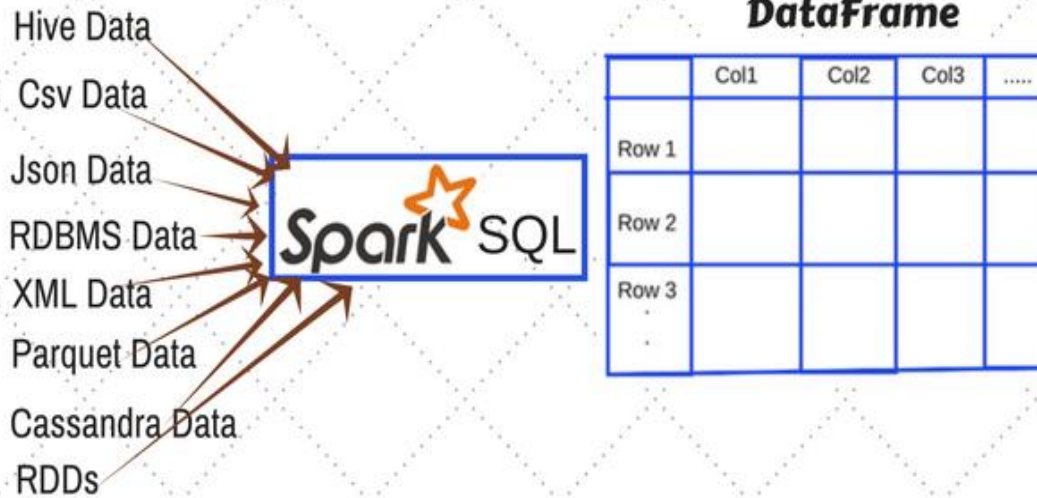
...

`(gene-8, (3.1, 5.1))`



Spark Data Abstractions: DataFrame

Ways to Create DataFrame in Spark



PySpark DataFrame Example-1

1. Import required libraries

```
from pyspark.sql import SparkSession
```

2. Create a SparkSession object

```
spark = SparkSession.builder.appName("demo").getOrCreate()
```

3. Create a sample DataFrame as df

```
data = [ ("Alice", "Sales", 45000),  
         ("Bob", "Engineering", 60000), ...]  
columns = ["name", "department", "salary"]  
df = spark.createDataFrame(data, columns)
```

4. Register the DataFrame as a Temporary SQL View

```
df.createOrReplaceTempView("employees")
```



PySpark DataFrame Example-1 continued...

5. Run an SQL query using spark.sql()

```
query_result = spark.sql("""
    SELECT
        department,
        COUNT(*) as total_employees,
        AVG(salary) as average_salary
    FROM employees
    GROUP BY department
    HAVING average_salary > 42000
    ORDER BY average_salary DESC
    """)
```

6. Display the results

```
query_result.show()
```



PySpark: create a DataFrame from a CSV File

1. Define your data

```
INPUT_PATH = "path/to/your/file.csv"
```

2. Read data from a CSV file with a header and create a DataFrame

```
df = spark.read.csv(  
    INPUT_PATH,  
    header=True,  
    inferSchema=True)
```

3. Find number of rows

```
df.count()
```

4. Show the first 10 rows of the DataFrame

```
df.show(10)
```



PySpark DataFrame Example-2 - 01

```
# This example introduces key intermediate concepts  
# like aggregation (groupBy), built-in column functions  
# (when, avg), and handling null values (na.fill).
```

```
# 1. Import required libraries
```

```
from pyspark.sql import SparkSession  
from pyspark.sql import functions as F
```

```
# 2. Initialize a SparkSession
```

```
spark = SparkSession.builder.appName("demo").getOrCreate()
```



PySpark DataFrame Example-2 - 02

```
# 3. Sample e-commerce transaction data
```

```
# (with some missing values)
```

```
data = [ ("User_A", "Electronics", 1200.0, "US"),  
          ("User_B", "Clothing", 45.50, "CA"),  
          ("User_C", "Electronics", None, "US"),  
          ("User_A", "Electronics", 150.0, "US"),  
          ("User_D", "Books", 22.00, "UK"),  
          ("User_B", "Clothing", 80.00, "CA"),  
          ("User_E", "Books", 15.75, None),  
        ]
```

```
Column_names = ["UserID", "Category", "Amount", "Country"]
```



PySpark DataFrame Example-2 - 03

4. Create a DataFrame

```
df = spark.createDataFrame(data, schema=column_names)
print("--- Raw Data ---")
df.show()
```

--- Raw Data ---

```
+-----+-----+-----+-----+
|UserID| Category|Amount|Country|
+-----+-----+-----+-----+
|User_A|Electronics|1200.0|US|
|User_B|Clothing|45.5|CA|
|User_C|Electronics|null|US|
|User_A|Electronics|150.0|US|
|User_D|Books|22.0|UK|
|User_B|Clothing|80.0|CA|
|User_E|Books|15.75|null|
+-----+-----+-----+-----+
```



PySpark DataFrame Example-2 - 04

```
# 5. Data Cleaning: Handle missing values
```

```
# Fill missing Amounts with 0.0, and
```

```
# Fill missing Countries with 'Unknown'
```

```
df_cleaned = df.na.fill({"Amount": 0.0, "Country": "Unknown"})
```

```
# 6. Data Transformation: Add a conditional "Customer_Segment" column
```

```
# High spenders (> $500), Medium spenders (> $0), and Inactive ($0)
```

```
df_transformed = df_cleaned.withColumn( "Segment",  
    F.when(F.col("Amount") > 500, "High")  
    .when(F.col("Amount") > 0, "Medium")  
    .otherwise("Inactive")  
)
```

```
# 7. Show the first 20 rows
```

```
print("--- Cleaned and Transformed Data ---")
```

```
df_transformed.show(20)
```



PySpark DataFrame Example-2 - 05

--- Cleaned and Transformed Data ---

```
+-----+-----+-----+-----+-----+
|UserID|  Category|Amount|Country|Segment |
+-----+-----+-----+-----+-----+
|User_A|Electronics|1200.0|    US|   High |
|User_B|  Clothing|  45.5|    CA| Medium |
|User_C|Electronics|   0.0|    US|Inactive|
|User_A|Electronics| 150.0|    US| Medium |
|User_D|      Books|  22.0|    UK| Medium |
|User_B|  Clothing|  80.0|    CA| Medium |
|User_E|      Books| 15.75|Unknown| Medium |
+-----+-----+-----+-----+-----+
```



PySpark DataFrame Example-2 - 06

8. Data Aggregation: Calculate metrics by Category

Total spend, average spend, and distinct customer count per category

```
df_metrics = df_transformed.groupBy("Category").agg(  
    F.sum("Amount").alias("Total_Revenue"),  
    F.round(F.avg("Amount").alias("Average_Spend"),  
    F.countDistinct("UserID").alias("Unique_Customers")  
)orderBy(F.desc("Total_Revenue"))
```

9. Show the first 20 rows

```
print("--- Aggregated Category Metrics ---")  
df_metrics.show(20)
```



PySpark DataFrame Example-2 - 07

--- Aggregated Category Metrics ---

Category	Total_Revenue	Average_Spend	Unique_Customers
Electronics	1350.0	450.0	2
Clothing	125.5	62.75	1
Books	37.75	18.88	2



3. Main Course Components

Serverless SQL Access to Big Data (15%)

- Amazon Athena
- Google BigQuery
- Snowflake



Amazon Athena: built from PrestoDB

Amazon Athena is a **serverless,**
interactive query service provided
by AWS that uses PrestoDB as its
underlying distributed SQL query
engine.



Amazon Athena: built from PrestoDB

- Interactive query service
- Serverless. Zero infrastructure. Zero administration.
- Put your data in S3
- Access your data by SQL
- Pay by query
- Fast performance



Amazon Athena

